Fiscal Year:	FY 2007	**Task Last Updated:**	FY 08/27/2008
PI Name:	Dulchavsky, Scott A. M.D., Ph.D.		
Project Title:	Minimally Invasive Diagnosis and Therapy of Microgravity Medical Contingencies		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Detroit	**State:**	MI
Zip Code:	48202-2608	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	07/01/2003	**End Date:**	06/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-SMS00301		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The diagnosis and management of acute health problems in space is problematic due to limited training of the Crew Medical Officer (CMO), human and environmental factors, and a lack of reference of the changes in anatomy, disease presentation, and therapy in micro-gravity. There is no planned radiological capability aboard the ISS further complicating medical diagnosis in space. Recent terrestrial investigations suggest expanded clinical applications of ultrasound and laparoscopy with miniature instrumentation which have applicability for aerospace medicine. This proposal will initially determine the diagnostic utility of ultrasound and/or micro-laparoscopy in select health contingencies with high potential mission impact. These diagnostic modalities will then be used to facilitate minimally invasive, definitive surgical therapy of selected contingencies in animal models in ground based and simulated micro-gravity scenarios. Optimal just-in-time training regimens and computer based refresher modules for non-physician CMOs to accomplish these tasks will be developed.		

Task Description:	The unique constraints imposed by training and equipment limitations and the space environment require the development of novel diagnostic and therapeutic strategies for crew member health problems including the expansion of ultrasound and mini-laparoscopy. Thoracic ultrasound, initially investigated by NASA as an alternative diagnostic modality for pneumothorax, has proven accuracy in terrestrial and micro-gravity applications and will have wide spread impact in acute care on Earth in the future. Although some of the techniques investigated in this proposal are appropriate only for a micro-gravity environment, the majority of the diagnostic and therapeutic algorithms are readily transferable to terrestrial medicine including rural and military applications. The expanded use of the diagnostic and training modalities described in this proposal, if verified, would provide a significant, clinically relevant advance in space medical capabilities with profound Earth-based ramifications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The just in time training algorithms developed in this proposal can be used to rapidly train non-medical personnel to perform complex medical diagnosis and treatment. This could be used in rural, military, or third world countries to deliver high-quality medical care. We have recently collaborated with a company, MEDIPHAN, which has video streaming technology to make ground based applications seamless. We will work in partnership to establish activities for sporting, military, and emergency applications.
Task Progress:	We have completed Pre-flight training and in-flight experiments on the ISS increment 10 and 11 crews and have completed post flight data acquisition on I 11 after landing. We have done numerous outreach activities with high school students and astronauts which have resulted in 2 publications to date. We are currently working with the Detroit Tigers, Detroit Red Wings, Detroit Lions, and United States Olympic Committee to modify our NSBRI developed training methodologies for use in these sporting arenas. We have completed preliminary ground investigations for microgravity flights on the new NASA research facility, and have completed a C9 flight comparing ultrasound with microlaparoscopy and have completed another set of experiments to evaluate cardiovascular ultrasound with or without the Russian Bracelet Device. We are completing data analysis and publications on the previous years' investigations. Our team delivered the plenary research presentation to the NSBRI/HRF Research meeting this year in Houston entitled Citius/Altius/Fortius reviewing the medical operations relevant experiments we have conducted to date. We are currenty in final analysis of our voluminous ultrasound and minimally invasive experiments prior to final publication.
Bibliography Type:	Description: (Last Updated: 07/24/2025)
Articles in Peer-reviewed Journals	Kwon D, Bouffard JA, van Holsbeeck M, Sargsyan AE, Hamilton DR, Melton SL, Dulchavsky SA. "Battling fire and ice: remote guidance ultrasound to diagnose injury on the International Space Station and the ice rink." Am J Surg. 2007 Mar;193(3):417-20. PMID: 17320547 , Mar-2007
Articles in Peer-reviewed Journals	Sargsyan AE, Dulchavsky AG, Adams J, Melton S, Hamilton DR, Dulchavsky SA. "Ultrasound detection of simulated intra-ocular foreign bodies by minimally trained personnel." Aviat Space Environ Med. 2008 Jan;79(1):58-61. PMID: 18225781 , Jan-2008
Articles in Peer-reviewed Journals	Benninger MS, McFarlin K, Hamilton DR, Rubinfeld I, Sargsyan AE, Melton SL, Moyhi M, McLaren PJ, Dulchavsky SA. "Ultrasonographic evaluation of sinusitis during microgravity in a novel animal model." Arch Otolaryngol Head Neck Surg. 2010 Nov;136(11):1094-8. http://dx.doi.org/10.1001/archoto.2010.196 ; PubMed PMID: 21079162 , Nov-2010